

GLUCOCORTICOIDS

- **Glucocorticoids act mainly on glucose metabolism.**
- **These are :—**
 - a. Cortisol
 - b. Corticosterone
 - c. Cortisone

SOURCE OF SECRETION

- **Mainly by:** Zona fasciculata of adrenal cortex.
- **Small quantity by:** Zona reticularis.

CHEMISTRY

- C_{21} steroids having 21 carbon atoms.
- **Half-life:**
 - 70 – 90 minutes (*Cortisol*)
 - 50 minutes (*Corticosterone*)

DAILY OUTPUT

- **Plasma level :—**
 - a. Cortisol: $13.9 \mu\text{g/dL}$
 - b. Corticosterone: $0.4 \mu\text{g/dL}$
- **Daily output :—**
 - a. Cortisol: $10 \mu\text{g}$
 - b. Corticosterone: $3 \mu\text{g}$

FUNCTIONS OF GLUCOCORTICIDS

- **Cortisol / Hydrocortisone** is more potent and has **95% of glucocorticoid activity**, corticosterone (**4%**) & cortisone has **1%**.

LIFE-PROTECTING HORMONE :—

Cortisol is a life-protecting hormone becoz it helps to withstand the stress and trauma in life.

1. ON CARBOHYDRATE METABOLISM

- **Glucocorticoids ↑se blood glucose level by 2 ways:**
 - a. **Promoting gluconeogenesis in liver from AA:**
↪ AA enter liver & get converted into glucose.
 - b. **Inhibiting uptake and utilization of glucose by peripheral cells:**
↪ (Action called **Anti-Insulin Action**).
- **Clinical significance:**
 - **Hypersecretion of glucocorticoids:** Results in hyperglycemia, glycosuria & **adrenal diabetes**.
 - **Hyposecretion:** Causes **hypoglycemia**.

2. ON PROTEIN METABOLISM

- **Glucocorticoids promote catabolism of proteins:**
 - ↓
 - a. ↓se in cellular proteins
 - b. ↑se in plasma level of AA
 - c. ↑se in protein content in liver.
- **Glucocorticoids cause catabolism of proteins by :—**
 - a. Releasing AA from body cells (*except liver cells*) into blood.
 - b. ↑sing uptake of AA by hepatic cells from blood.

Hypersecretion of glucocorticoids (*excess catabolism of proteins*):

↓

Result in **muscular wasting & -ve nitrogen balance**.

3. ON FAT METABOLISM

- **Glucocorticoids cause mobilization and redistribution of fats:**
 - a. Mobilization of FA from adipose tissue.
 - b. ↑sing the concentration of FA in blood.
 - c. ↑sing the utilization of fat for energy.
 - d. Glucocorticoids ↓se utilization of glucose:
 - ↓
 - **Ketogenic effect of glucocorticoids**
 - ↓
 - Glucose → FA → Large amount of ketone bodies
- **Hypersecretion of glucocorticoids causes:**
 - ↪ **Obesity** (*deposition of fat at abdomen, chest, face*).

4. ON WATER METABOLISM

- **Glucocorticoids maintain water balance**, by accelerating excretion of water.
- **Adrenal insufficiency:** Causes **water retention and water intoxication** after intake of large quantity of H_2O .

5. ON MINERAL METABOLISM

- **Glucocorticoids enhance retention of Na^+** and to lesser extent, ↑se the excretion of K^+ .
- **Hypersecretion of glucocorticoids → Causes:**
 - a. Edema
 - b. Hypertension
 - c. Hypokalemia
 - d. Muscular weakness
- **Glucocorticoids ↓se the blood Ca^{2+}** by inhibiting its absorption from intestine & ↑sing excretion through urine.

6. ON BONE

- **Stimulate bone resorption** (*osteoclastic activity*) and **inhibit bone formation and mineralization** (*osteoblastic activity*).
- **Hypersecretion: Osteoporosis** occurs.

7. ON MUSCLES

- ↑**se** catabolism of proteins in muscle.
- **Hypersecretion causes:** Muscular weakness due to loss of proteins.

8. ON BLOOD CELLS

- **Glucocorticoids** ↓**se no. of circulating eosinophils** by ↑sing the destruction of eosinophils in reticuloendothelial cells.
- **Also** ↓**se** basophils & lymphocytes.
- ↑**se** neutrophils, RBC & platelets.

9. ON VASCULAR RESPONSE

- **Glucocorticoids essential for constrictor action** of adrenaline and noradrenaline.
- **In adrenal insufficiency:** Blood vessels fail to respond to adrenaline and noradrenaline:
 - ↓
 - Leads to **vascular collapse**.

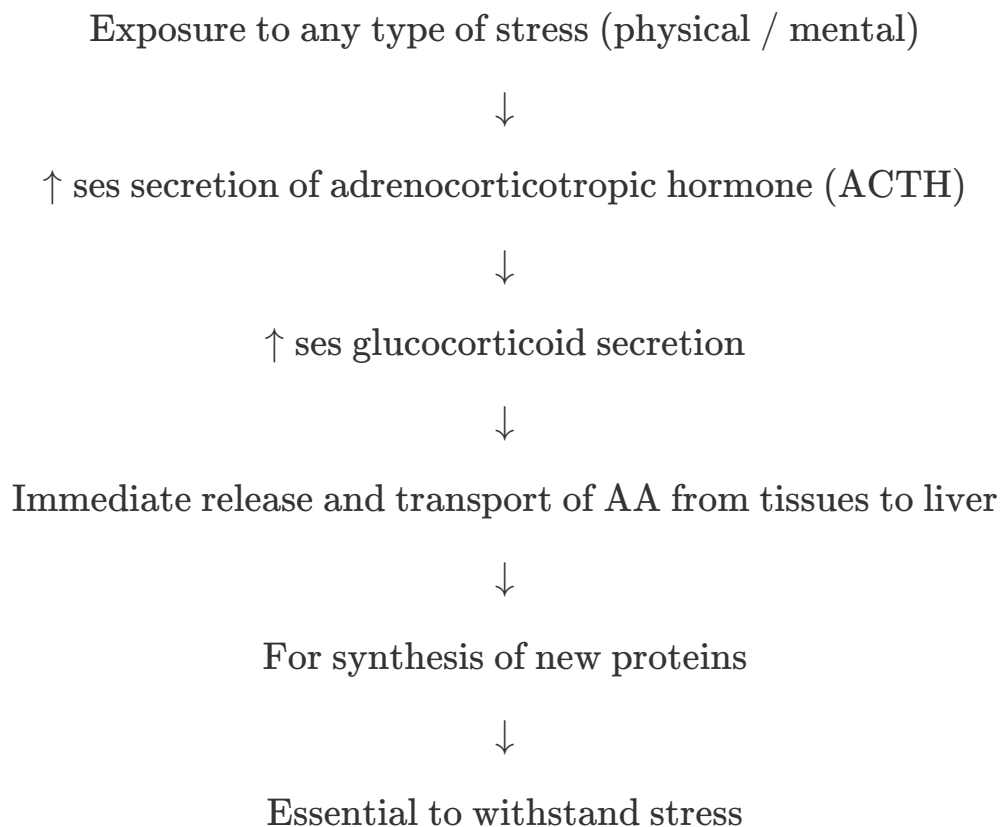
10. ON CNS

- **Essential for normal functioning of nervous system.**
- **Insufficiency causes:** Personality changes like irritating & lack of concentration.

11. PERMISSIVE ACTION OF GLUCOCORTICOIDS

- Refers to execution of actions of some hormones only in presence of glucocorticoids:
 - a. Calorigenic effect of glucagon
 - b. Lipolytic effect of catecholamines
 - c. Vascular effects of catecholamines
 - d. Bronchodilator effect of catecholamines

12. ON RESISTANCE OF STRESS



13. ANTI-INFLAMMATORY EFFECTS

14. ANTI-ALLERGIC ACTIONS

15. IMMUNOSUPPRESSIVE EFFECTS

REGULATION OF SECRETION

